

Criterion B: Design

1. Basic system structure

The basic system structure will be composed of four components:

1. A website that the client may interact with for inputting, outputting using HTML, CSS and JavaScript.
2. A third-party livestreaming program such as OBS that allows the streamer to broadcast live video to the server.
3. A server that is able to manage user requests, live streams and chat messages of the users. Uses Rust as the programming language with a HTTP framework.
4. An SQLite database to store user credentials and sessions.

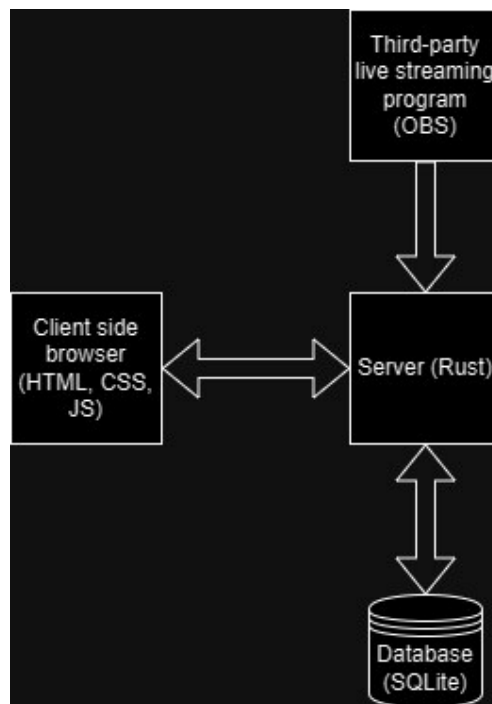


Figure B.1: Basic system diagram

2. Functionality

The flowchart as shown in Figure B.2 presents how the system is able to accomplish its success criteria.

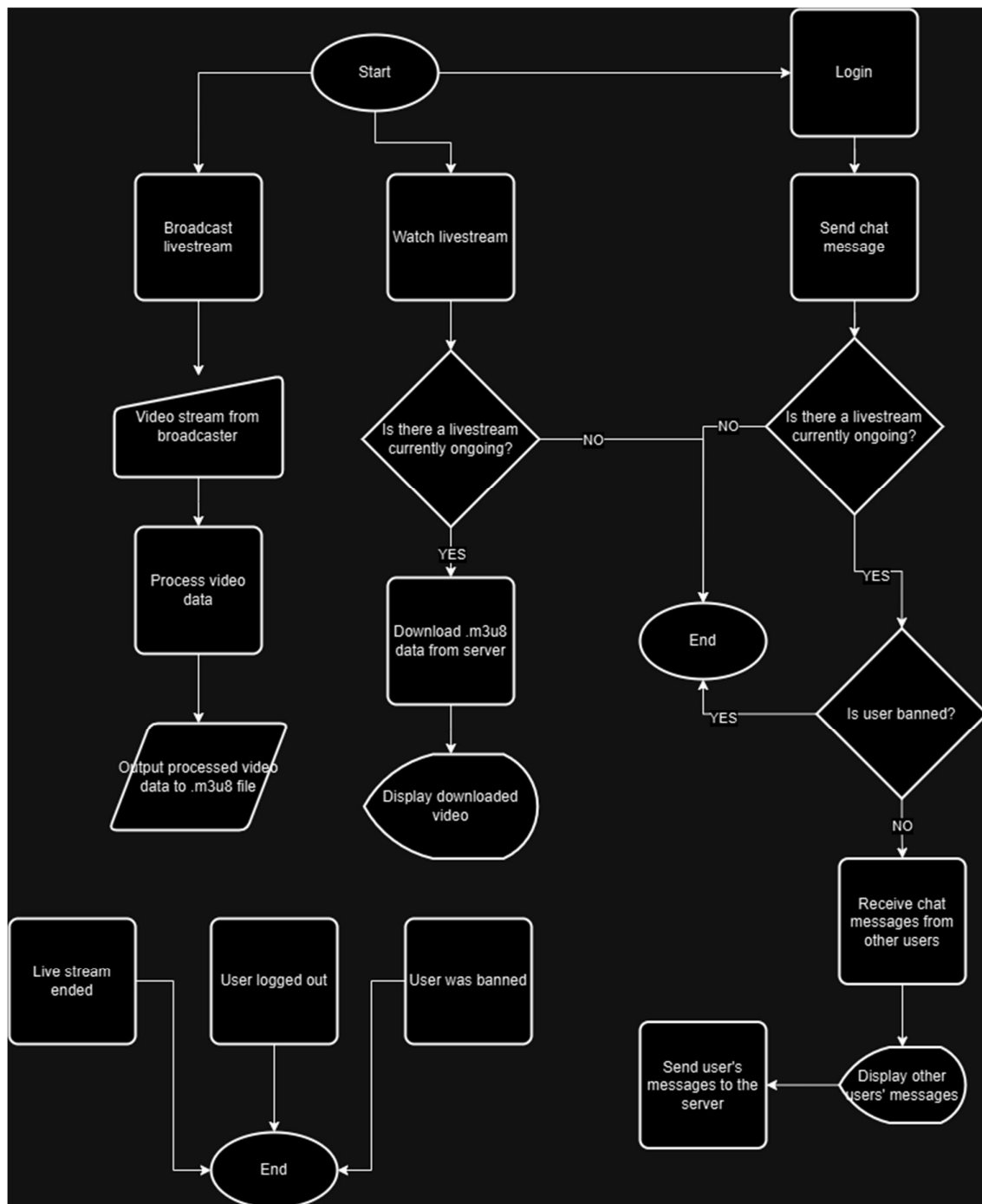


Figure B.2: Main functionality flowchart

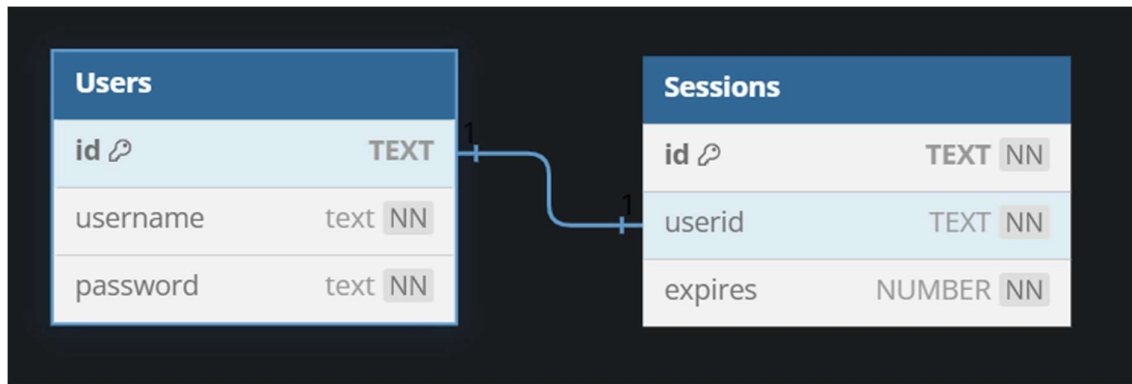


Figure B.3: Database entity relationship diagram (ERD)

3. Pages

Page	Route	Description
Main page (Criterion 2, 4)	/	The main site which contains the actual livestream together with the chat. The user may access the login page from the main page.
Login (Criterion 3)	/login	The page which contains the login form so that the user may access their account.
Register (Criterion 3)	/register	The page which contains the registration form which allows the user to create a new account in order to use the chat.
Administration (main page)	/admin	A special page protected only by a password containing administrative tools and settings. From this page the administrator may change the stream key and manage the livestream's title and description.
Administration (Custom CSS) (Criterion 6, 7)	/admin/css	A sub-page of the administration page where the administrator may change the CSS

		stylesheet of the main page.
Stream input	/stream	A page that contains the constantly updated m3u8 file which will be fetched by the browser in order to be displayed to the end user.

4. Page designs

4.1. Main page

This is the main page which will display the livestream, chat, title and description of the livestream. The page may have a different layout if the customer chooses to apply custom CSS.

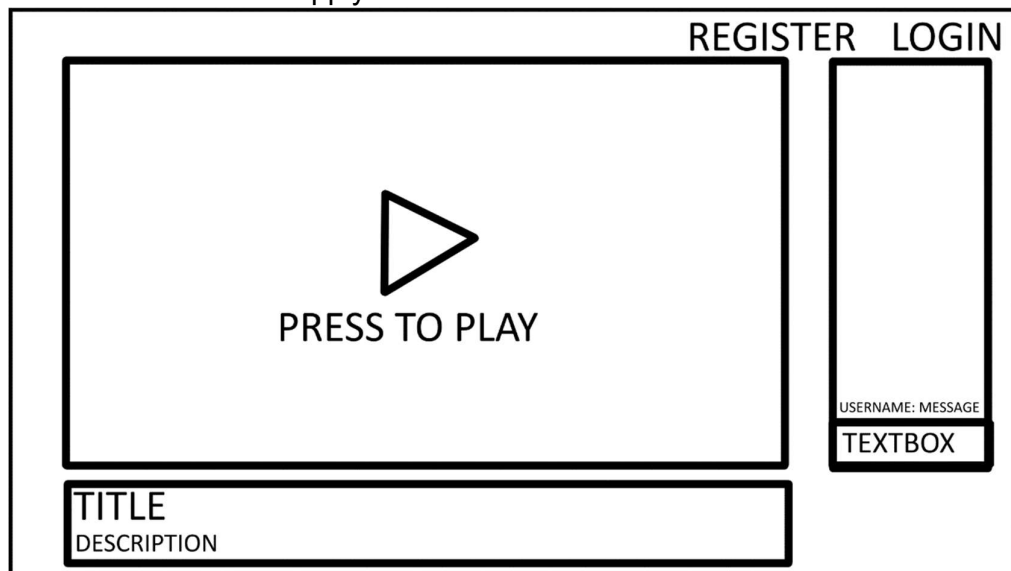


Figure B.4: Main page

4.2. Main administration page

This is the main administration page from which the customer will be able to manage the entire website and access other administration components, such

as accounts or CSS.

ADMINISTRATION PAGE (MAIN)		MAIN	CSS	ACCOUNTS
STREAM KEY (READ-ONLY)		COPY RESET		
STREAM TITLE		SET		
DESCRIPTION		SET CLEAR		

Figure B.5: Main administration page

4.3. CSS administration page

Below is the design sketch of the CSS administration page, using which the customer will be able to modify the layout of the main page as mentioned in criterion 7.

ADMINISTRATION PAGE (CSS)		MAIN	CSS	ACCOUNTS
CSS CODE		SET		
		RESET TO DEFAULT		
		HELP (CSS GUIDE)		

Figure B.6: CSS administration page

4.4. Account registration page

The drawing below presents the account registration page. In order to register, the user needs to provide the username and password they wish to have.

WATCH

LOGIN

Username:

Password:

Confirm password:

REGISTER BUTTON

Figure B.7: Registration page

4.5. Account login page

The sketch below shows the account login page. If a user choses to login they must provide their username and password.

WATCH

REGISTER

Username:

Password:

LOGIN BUTTON

Figure B.8: Login page

5. Testing plan

Test condition	Expected outcome	Relevant success criteria
Login with incorrect data.	Website refuses to login.	3
Login with correct data.	Website redirects to main page with user being logged in.	3
Register an account.	Website creates account and logs in user.	3

Start streaming from OBS using incorrect stream key.	Server refuses to show broadcast to viewers.	1
Start streaming from OBS using correct stream key..	Server accepts the stream and re-transmits it to the viewers.	1, 2
Send message in the chat using a newly registered account.	Message is displayed in the chat for other users.	4
As website administrator, ban another account.	Banned account loses the ability to send messages.	5
Change stream title and description.	Stream title changes when main site is refreshed.	6
Insert custom CSS.	Main stream site style and look changes when refreshed.	7